

In Silico Toxicological Qualification of Two Vancomycin B Impurities by 3D-QSAR / Structure-Based Read-Across

Impurities RRT 0.75 [(26R)-DMe-DeLI] and RRT 0.87 [(7S,26R)-Vancomycin B]

Test items	Vancomycin B (API) and impurities RRT 0.75 and RRT 0.87
In silico methods	Toxicophore/pharmacophore profiling; RDKit PAINS/Brenk/NIH alerts; physicochemical & 3D-shape descriptors; 3D conformer generation & MCS/O3A shape read-across; in-house 12-endpoint Tox21 statistical QSAR (RF, 5-fold CV, AD); ICH M7 two-methodology mutagenicity screen (Benigni–Bossi expert rulebase + Ames QSAR); supporting logD/pKa QSAR
Toxicological anchor	Parent API 90-day i.v. dog study, NOAEL 75/100 mg/kg/day (~2–3× clinical dose)
Date	2026-07-30

IMPORTANT — This report is a computational screening and structure-based read-across assessment intended to support expert toxicological review. It is not, by itself, a regulatory toxicology conclusion or a substitute for expert sign-off or, where required, experimental testing (see §8).

1. Executive summary

Two related substances of the glycopeptide antibiotic vancomycin B require in silico toxicological qualification: **RRT 0.87 = (7S,26R)-vancomycin B**, a heat/low-water-activity degradation product, and **RRT 0.75 = (26R)-DMe-DeLI**, a fermentation-derived residue-1 variant. Because the differences are largely stereochemical/constitutional and not resolvable by conventional 2D read-across, a three-dimensional, structure-based approach was applied to the parent and both daughters.

Principal conclusion. Neither impurity introduces a new toxicophore or a new DNA-reactive (mutagenic) structural alert absent from the parent API. RRT 0.87 is a pure stereoisomer sharing all 101 heavy atoms and the complete functional-group inventory of the parent (differing only in configuration at 2 of 18 stereocentres). RRT 0.75 shares 99 of ~100 heavy atoms; its only constitutional change is a secondary (N-methyl) → primary aliphatic amine at residue 1 — a group class already present in the parent (vancosamine) — with unchanged lipophilicity (logD 0.42) and near-identical basicity (pKa ≈ 8.1 vs 8.2).

A statistical 12-endpoint Tox21 QSAR (mean 5-fold CV-AUC 0.827) reinforces this: the parent is itself a Tox21 training compound (measured profile essentially clean — only a weak NR-AR signal), RRT 0.87 is fingerprint-identical to the parent ($\Delta \equiv 0$ at all endpoints), and RRT 0.75 differs from the parent by ≤ 0.02 at every endpoint. An ICH M7 two-methodology mutagenicity screen (Benigni–Bossa expert rulebase + Ames QSAR, AUC 0.881) returns a concordant **negative / Class 5** for all three. On a weight-of-evidence basis both impurities qualify for read-across to the parent, whose safety is anchored by the 91-day i.v. dog study (NOAEL 75/100 mg/kg/day) and extensive clinical use, provided each stays within the Ph. Eur. related-substances limits.

Question	RRT 0.87	RRT 0.75	Basis
New toxicophore vs parent?	NO	NO	Identical FG inventory (0.87); iso-functional amine swap (0.75)
New mutagenic (M7) alert?	NO	NO	Both M7 methodologies negative → Class 5 (§5.7)
Read-across to parent justified?	YES	YES	≥98% shared skeleton; conserved core toxicophores

2. Purpose, scope and regulatory framework

The objective is a toxicological in silico qualification of impurities RRT 0.75 and RRT 0.87, delivering: (i) a 3D-QSAR / structure-based assessment of the parent and each daughter; (ii) identification/assessment of any additional toxophore present in a daughter but absent from the parent; (iii) a documented read-across justification; and (iv) this regulatory-style report.

Framework. Non-mutagenic impurity qualification follows ICH Q3A(R2)/Q3B(R2). Mutagenic potential is screened per ICH M7(R2), which anticipates two complementary (Q)SAR methodologies (expert rule-based + statistical) plus expert review. In silico model use is framed by the OECD (Q)SAR validation principles. Because vancomycin B is a large (MW $\approx$ 1449) flexible glycopeptide outside the applicability domain of many small-molecule tox models, the assessment is weighted toward mechanistic, structure- and shape-based read-across, with statistical models used where their domain is respected (and flagged where not).

3. Test items and structural relationships

Structures were taken from the sponsor information (IUPAC/SMILES/InChI) and independently validated with RDKit; computed formulae and exact masses reproduce the source document.

Attribute	Parent API	RRT 0.87	RRT 0.75
Xellia name	Vancomycin B	(7S,26R)-Vancomycin B	Vancomycin B impurity (26R)-DMe-DeLI
Molecular formula	C ₆₆ H ₇₅ Cl ₂ N ₉ O ₂₄	C ₆₆ H ₇₅ Cl ₂ N ₉ O ₂₄	C ₆₅ H ₇₃ Cl ₂ N ₉ O ₂₄
Monoisotopic MW	1447.43	1447.43	1433.41
Defined stereocentres	18	18	19
Origin	Fermentation (API)	Degradation (low water activity + heat)	Fermentation
Relationship to API	—	Diastereomer (C7 & C26 epimer)	Residue-1 variant (N-desMe; Leu → Ile) + C26 epimer

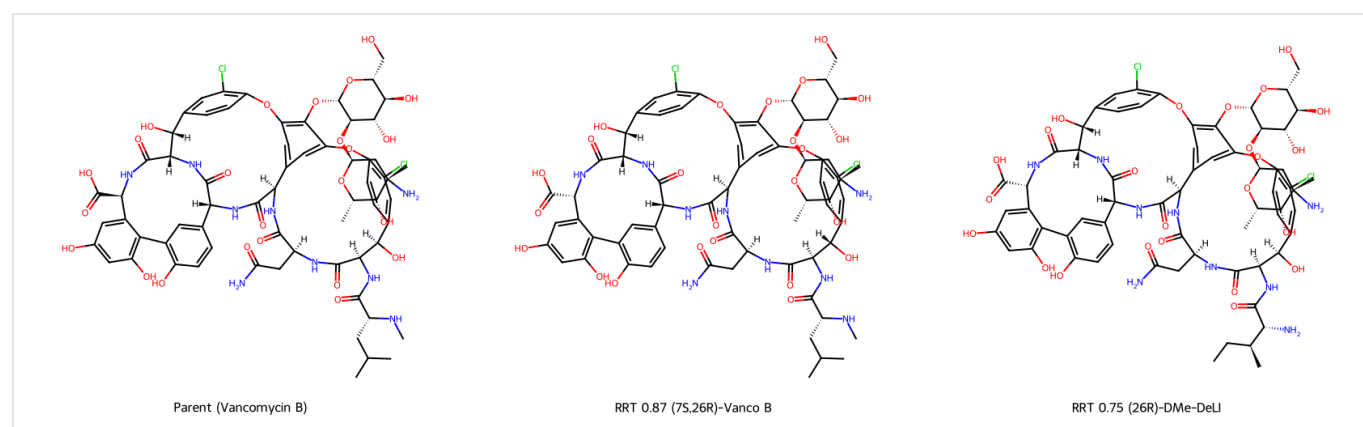


Figure 1. 2D structures of the parent API (left) and the two impurities. RRT 0.87 is drawn identically to the parent because it differs only in stereochemistry; RRT 0.75 differs at N-terminal residue 1 (free primary amine + isoleucine side chain instead of N-methyl-leucine).

RRT 0.87 — (7S,26R)-vancomycin B. Same molecular formula as the API; configuration inverted at C7 (benzylic carbinol of residue 3) and C26 (a peptide α -carbon) — consistent with thermally-driven epimerisation under low water activity. No atoms added or removed.

RRT 0.75 — (26R)-DMe-DeLI. Formula C₆₅H₇₃Cl₂N₉O₂₄ (one CH₂ less). Residue 1 changes from N-methyl-D-leucine to des-N-methyl-isoleucine (secondary → primary amine; isobutyl → sec-butyl), with epimerisation at C26. All other residues, both sugars, both aryl chlorides and all phenols are retained.



Parent: N-methyl-D-leucine (secondary amine).

RRT 0.75: des-methyl isoleucine (primary amine).

Figure 2. The only site of constitutional change between RRT 0.75 and the API.

4. Computational methods

4.1 Toxicophore / structural-alert profiling

Each structure was profiled for the vancomycin pharmacophore (heptapeptide aglycone D-Ala-D-Ala cleft, H-bonding network, vancosamine–glucose disaccharide) and the recognised toxicophores (phenolic/aromatic rings → renal accumulation; basic amine centres → histamine/Red-Man; high MW/hydrophilicity). A quantitative functional-group inventory (SMARTS) was differenced against the parent. Liabilities were screened with RDKit PAINS(A/B/C), Brenk and NIH catalogues and reviewed against ICH M7 DNA-reactive alert classes.

4.2 Physicochemical & 3D descriptors; 3D shape read-across

RDKit descriptors (MW, cLogP, TPSA, HBD/HBA, rot. bonds, QED, Lipinski/Veber); Inductive Bio logD/pKa QSAR. ETKDGV3 conformers, MMFF-minimised; 3D shape descriptors (RoG, asphericity, sphericity, eccentricity, NPR1/2). 3D read-across quantified by MCS core-restricted alignment (RMSD) and Crippen Open3DAlign / shape-Tanimoto. A formal CoMFA/CoMSIA 3D-QSAR is not built (see §8).

4.3 Statistical QSAR — Tox21 and Ames

A 12-endpoint Tox21 QSAR (class-balanced RandomForest on 2048-bit Morgan fingerprints; public MoleculeNet Tox21; 5-fold OOF AUC) and an Ames QSAR (Hansen benchmark N≈6500) were trained and applied with applicability-domain checks. Parent–impurity differences are the interpretable read-out.

5. Results

5.1 Parent (Vancomycin B) — baseline toxicophore profile

The parent presents the expected glycopeptide profile: 6 phenolic OH, 5 aromatic rings, 2 aryl chlorides, 3 aryl ethers, 2 benzylic alcohols, 1 carboxylic acid, 1 primary amide, 6 secondary amides, 1 primary amine (vancosamine), 1 secondary N-methyl amine. It carries no PAINS/Brenk alert and no DNA-reactive alert; the single NIH flag (≥ 7 aliphatic OH) is a polarity descriptor. This defines the read-across baseline.

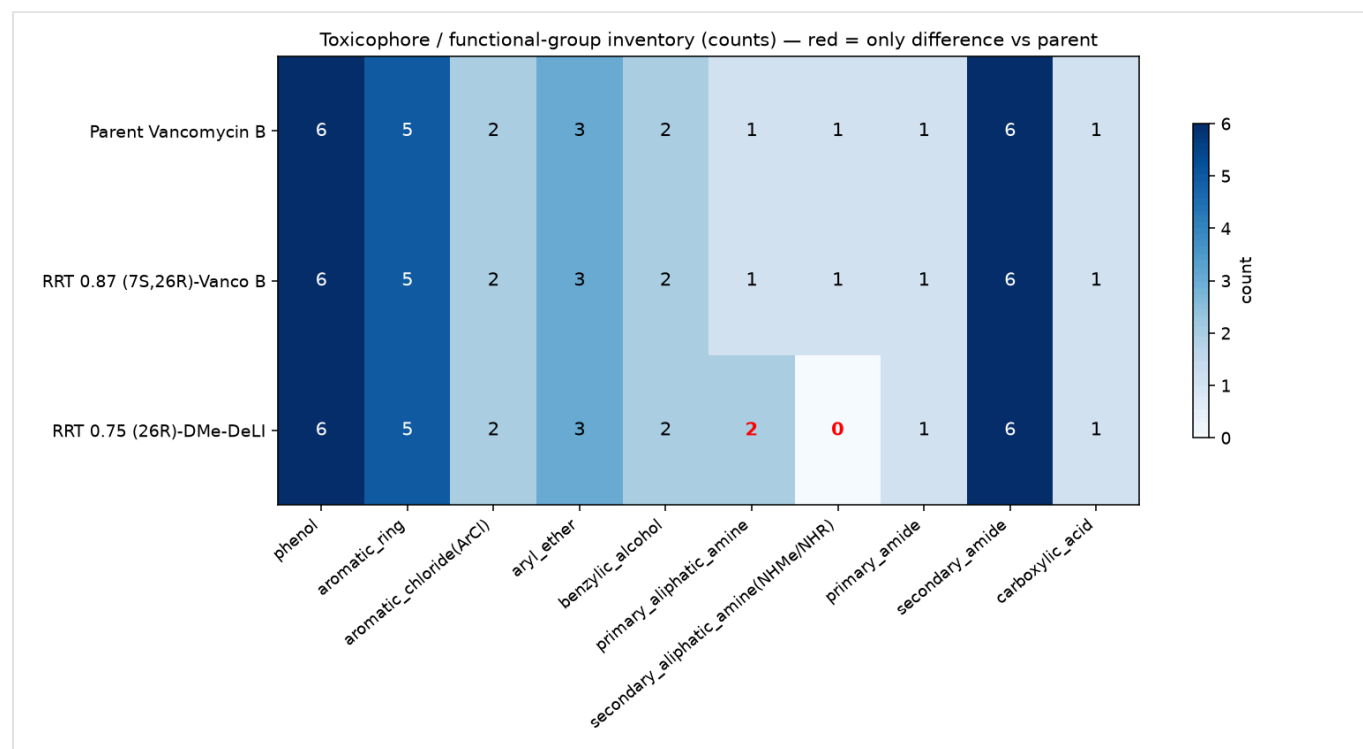


Figure 3. Toxicophore / functional-group inventory (counts). Parent and RRT 0.87 are identical across every category; RRT 0.75 differs only in the amine columns (red) — one secondary (N-methyl) amine replaced by one primary amine.

5.2 RRT 0.87 — (7S,26R)-vancomycin B

Differing from the API only in configuration, its 2D structure, functional-group inventory and alert profile are identical to the parent (ECFP4/MACCS Tanimoto = 1.00; 101/101 shared heavy atoms; 0 alerts). **No additional toxicophore.** Conventional 2D read-across cannot distinguish it from the API — which is why a 3D/mechanistic assessment is required. 3D shape descriptors are near-congruent (RoG 7.45 vs 7.50 Å; NPR1 0.529 vs 0.528). Stereochemical inversion can modulate target binding (efficacy) but creates no new reactive/DNA-reactive functionality.

5.3 RRT 0.75 — (26R)-DMe-DeLI

Retains the entire aglycone, both sugars, both aryl chlorides, all phenols, the carboxyl and the primary amide; 99/~100 heavy atoms shared (ECFP4 0.86 / MACCS 0.94). The single change (secondary N-methyl → primary amine) is **not a new functional-group class** (parent already carries a primary amine) and is **not a DNA-reactive alert**. logD unchanged (0.42 vs 0.42), most-basic pKa near-identical (8.08 vs 8.21) — the basic-amine/Red-Man liability is conserved, not amplified. TPSA +14 Å²; 3D shape close to parent (RoG 7.46 Å).

5.4 Differential toxicophore analysis — is any new toxicophore present?

Potential toxicophore	RRT 0.87 vs parent	RRT 0.75 vs parent
Phenolic / aromatic rings	Identical (6 phenol, 5 aryl)	Identical (6 phenol, 5 aryl)
Basic amine centres (histamine/Red-Man)	Identical	Conserved: sec → prim amine, iso-basic (pKa 8.1); not amplified
Aryl chlorides	Identical (2)	Identical (2)
DNA-reactive alert (arom. amine, nitro, epoxide, Michael acceptor, alkyl halide...)	None (as parent)	None (as parent)
PAINS / Brenk liability	None	None
NET: new toxicophore introduced?	NO	NO

The only element warranting comment is the residue-1 amine in RRT 0.75 — a modification of an existing toxicophore class (secondary → primary amine of equal basicity), not a new one. No additional toxicophore is identified in either daughter.

5.5 3D-QSAR / shape read-across metrics

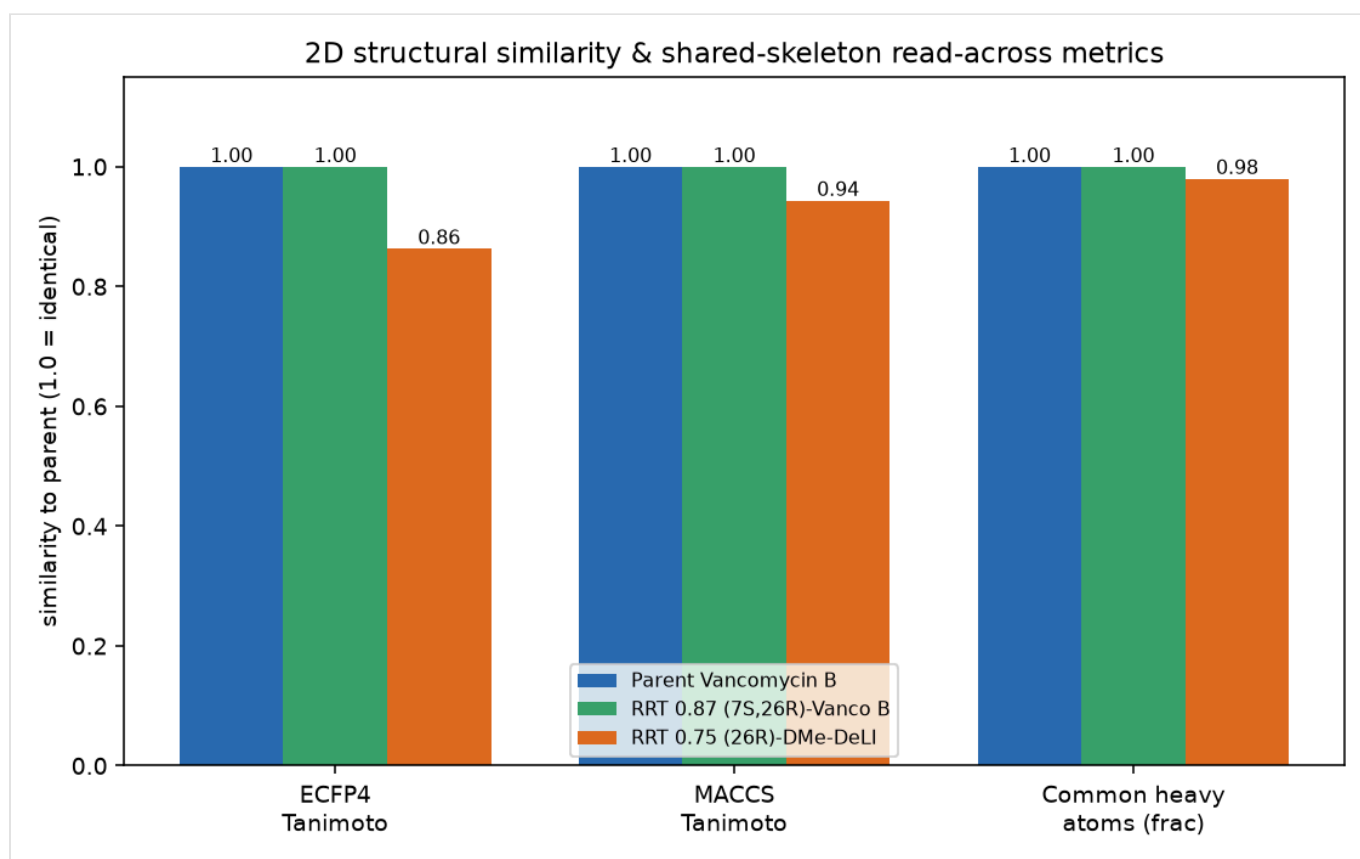


Figure 4. 2D similarity & shared-skeleton read-across. RRT 0.87 is fingerprint-identical; RRT 0.75 shares ≥98% of the heavy-atom skeleton.

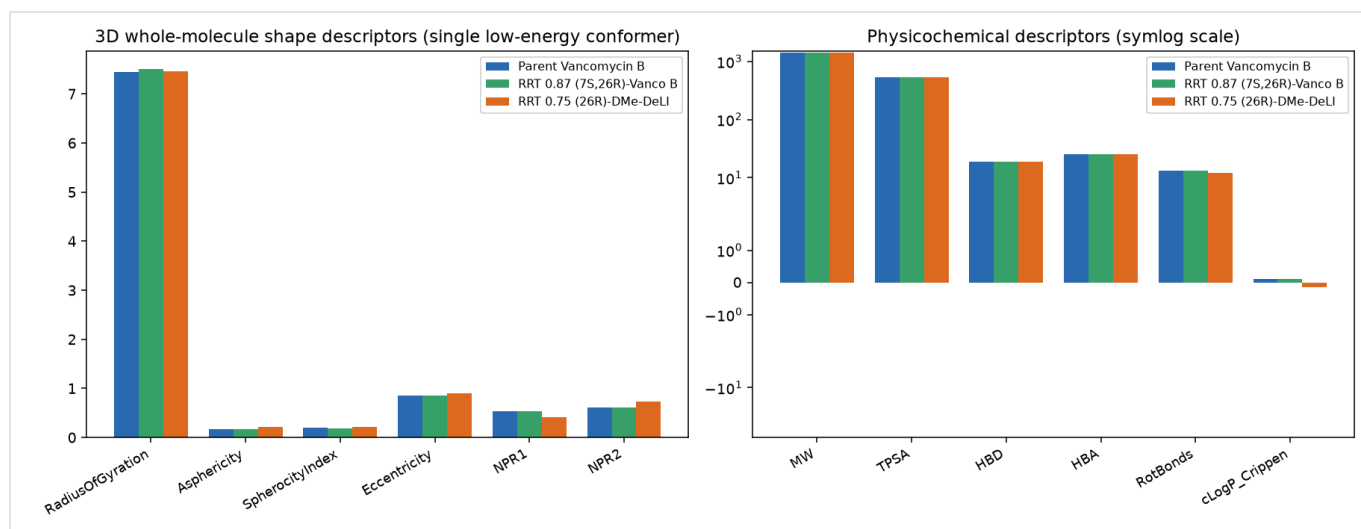


Figure 5. 3D whole-molecule shape descriptors (left) and physicochemical descriptors (right, symlog). The three occupy essentially the same shape/property space.

5.6 Statistical (2D) QSAR — 12-endpoint Tox21 profiling

A 12-endpoint Tox21 QSAR (class-balanced RandomForest on 2048-bit Morgan fingerprints; public MoleculeNet Tox21, $n=7823$) was trained: mean 5-fold OOF ROC-AUC = 0.827 (range 0.723–0.892), then each molecule scored with an applicability-domain check.

Applicability domain — a favourable finding. Vancomycin is itself in the Tox21 collection (training compound TOX25354; nearest-neighbour Tanimoto = 1.00 to the parent and to RRT 0.87). The QSAR is therefore in-domain for the parent; RRT 0.75 sits close by (max Tanimoto 0.863).

Measured vancomycin profile (ground truth): active only at NR-AR (weak), inactive at every other tested endpoint (incl. genotoxicity SR-ATAD5, SR-p53). This is the toxicological baseline.

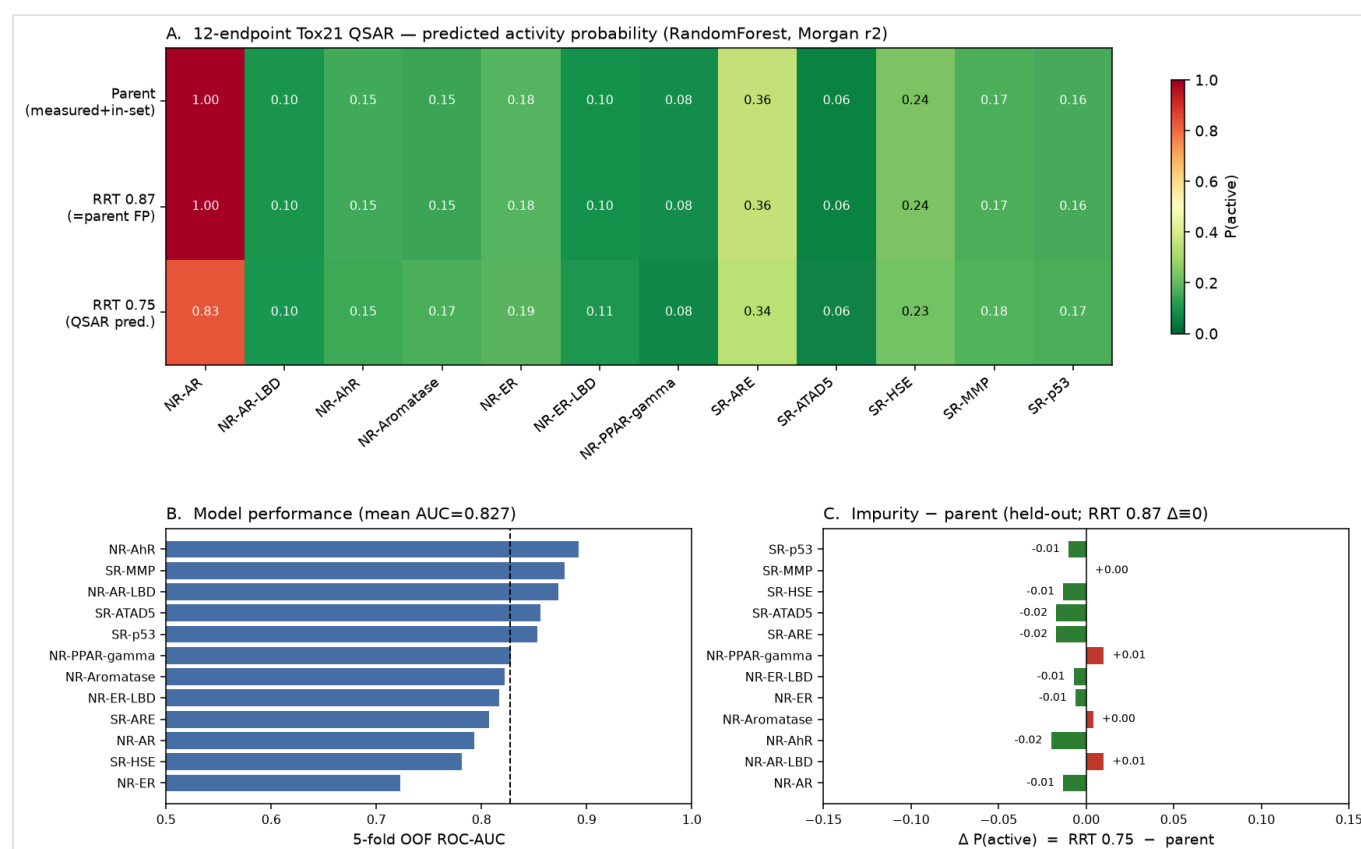


Figure 6. In-house 12-endpoint Tox21 QSAR. (A) Predicted probabilities — parent and RRT 0.87 rows identical (identical fingerprints). (B) 5-fold OOF AUC per endpoint (mean 0.827). (C) RRT 0.75 – held-out parent Δ — every endpoint within ± 0.02 ; RRT 0.87 $\Delta \equiv 0$ by construction.

Impurity vs parent. RRT 0.87 returns identical probabilities at all 12 endpoints ($\Delta \equiv 0$): a concrete demonstration that a 2D-QSAR is structurally blind to this stereoisomer (hence the 3D read-across carries the weight). For RRT 0.75, every endpoint lies within ± 0.02 of the held-out parent (max $|\Delta| = 0.02$), none crossing the activity threshold — including the genotoxicity-associated SR-ATAD5 and SR-p53.

Endpoint	CV-AUC	Measured (parent)	Held-out parent P	RRT 0.75 P (Δ)
NR-AR (Androgen receptor (full))	0.793	active	0.219	0.206 (-0.013)
NR-AR-LBD (Androgen receptor (LBD))	0.873	inactive	0.220	0.230 (+0.010)
NR-AhR (Aryl hydrocarbon receptor)	0.892	inactive	0.200	0.180 (-0.020)
NR-Aromatase (Aromatase (CYP19A1))	0.822	inactive	0.223	0.227 (+0.004)
NR-ER (Estrogen receptor (full))	0.723	inactive	0.306	0.300 (-0.006)
NR-ER-LBD (Estrogen receptor (LBD))	0.817	inactive	0.160	0.153 (-0.007)
NR-PPAR-gamma (PPAR-gamma)	0.828	inactive	0.163	0.173 (+0.010)
SR-ARE (Oxidative-stress response (ARE))	0.807	not tested	0.357	0.340 (-0.017)
SR-ATAD5 (Genotoxicity / DNA damage (ATAD5))	0.856	inactive	0.150	0.133 (-0.017)
SR-HSE (Heat-shock response)	0.781	not tested	0.240	0.227 (-0.013)
SR-MMP (Mitochondrial membrane potential)	0.879	inactive	0.260	0.260 (+0.000)
SR-p53 (p53 DNA-damage response)	0.853	inactive	0.303	0.293 (-0.010)

All predicted probabilities are below the 0.5 threshold (parent NR-AR is a measured weak active the model under-calls — expected at that endpoint's lower AUC; the measured label governs). RRT 0.87 omitted from the table: identical fingerprint $\Rightarrow$ identical predictions ($\Delta \equiv 0$).

5.7 ICH M7 mutagenicity assessment — two complementary (Q)SAR methodologies

ICH M7(R2) requires potential bacterial (Ames) mutagenicity to be assessed by two complementary methodologies — one **expert rule-based** and one **statistical** — with any positive resolved by expert review. Both were applied:

- **Methodology 1 (expert rule-based).** A Benigni–Bossa structural-alert rulebase for mutagenicity/genotoxic carcinogenicity (29 alert classes: aromatic amines, nitro/nitroso, azo/azoxy, N-nitroso, epoxides/aziridines, Michael acceptors, acyl/alkyl halides, hydrazines, aldehydes, quinones, β -lactams, ...) — the public scientific rulebase underlying Derek-type systems (Toxtree/VEGA).
- **Methodology 2 (statistical).** An Ames QSAR (class-balanced RandomForest, 2048-bit Morgan) trained on the Hansen benchmark (N = 6506; 54% mutagens); 5-fold OOF AUC = 0.881; decision threshold 0.52.

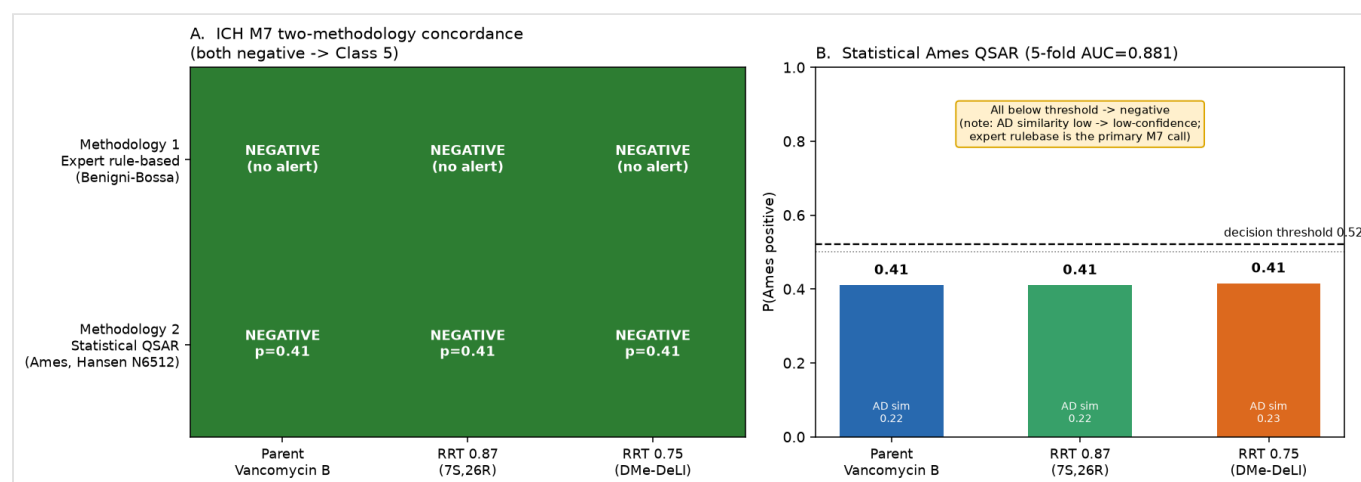


Figure 7. ICH M7 two-methodology assessment. (A) Concordance matrix — both methodologies NEGATIVE for all three molecules. (B) Statistical Ames probabilities, all below threshold; amber note flags the low applicability-domain similarity, which is why the domain-independent expert rulebase is the primary M7 call.

Molecule	Expert rule-based	Statistical Ames QSAR	M7
Parent (Vancomycin B)	negative (no structural alert)	negative (p=0.41, AD 0.22)	5
RRT 0.87 (7S,26R)-Vanco B	negative (no structural alert)	negative (p=0.41, AD 0.22)	5
RRT 0.75 (26R)-DMe-DeLI	negative (no structural alert)	negative (p=0.41, AD 0.23)	5

Outcome. The two methodologies are concordant and negative for the parent and both impurities: no mutagenicity structural alert, and the statistical Ames prediction below threshold in every case. Under ICH M7 this corresponds to **Class 5 (no structural alert)** — treated as non-mutagenic and controlled as ordinary ICH Q3A related substances, with no dedicated Ames study required. For transparency: (i) RRT 0.87 is fingerprint-identical to the parent, so both methodologies return identical results by construction; (ii) these glycopeptides lie outside the Ames model's applicability domain (nearest-neighbour Tanimoto $\approx$ 0.22), so the statistical negative is low-confidence and the domain-independent expert rulebase — unambiguously alert-free — is the primary determinant, corroborated by the parent's experimentally inactive Tox21 DNA-damage endpoints (§5.6). A licensed Derek/Sarah Nexus run is expected to reproduce this alert-free, negative call.

6. Read-across justification

Source (analogue): the parent API, vancomycin B (full toxicological package incl. 91-day i.v. dog study, NOAEL 75/100 mg/kg/day, plus decades of clinical exposure). **Targets:** RRT 0.87 and RRT 0.75.

- **Structural similarity.** RRT 0.87 shares 100% of the heavy-atom skeleton and functional groups (configurational isomer). RRT 0.75 shares $\geq 98\%$ (99/100), with a single localised, non-alerting modification at a peptide N-terminus.
- **Common mechanism / conserved toxicophores.** All phenolic, aromatic, aryl-chloride, amide, carboxyl and basic-amine features are conserved. No new reactive centre.
- **Metabolic / physicochemical congruence.** Vancomycin is renally cleared largely unchanged; near-identical size, polarity, logD (0.42) and pKa (≈ 8.1) predict the same ADME/disposition.
- **No mutagenic concern.** Both ICH M7 methodologies negative (§5.7); QSAR shows no activation at SR-ATAD5/SR-p53 ($\Delta \leq 0.02$).
- **Acknowledged uncertainty (stereochemistry).** Stereoisomers can differ in target binding — bearing on potency/efficacy, not on de novo toxicity; the 3D analysis specifically addresses this dimension.

7. Qualification conclusion

On a weight-of-evidence basis the in silico assessment supports toxicological qualification of both impurities by read-across to the parent:

- **RRT 0.87 [(7S,26R)-vancomycin B]** — a pure stereoisomer with no new toxicophore and no new mutagenic alert; read-across fully justified.
- **RRT 0.75 [(26R)-DMe-DeLI]** — a single, non-alerting, iso-functional residue-1 modification (plus C26 epimerisation) with no new toxicophore and no new mutagenic alert; read-across justified.

Provided both remain within the Ph. Eur. related-substances limits, the parent's NOAEL (75/100 mg/kg/day, $\approx 2\text{--}3\times$ clinical) and clinical record provide adequate toxicological cover. Formal sign-off should be issued by a qualified toxicologist, with the mutagenic dimension documented under the company's ICH M7 two-(Q)SAR-methodology workflow.

8. Applicability domain, honesty statement and limitations

- **Nature of the "3D-QSAR".** No congeneric training set with measured toxicity endpoints exists for these compounds, so a formal field-based 3D-QSAR (CoMFA/CoMSIA) predictive model was not — and cannot legitimately be — built. What is reported is 3D-structure/shape read-across plus mechanistic toxicophore profiling.
- **Statistical QSAR — scope of validity.** The 12-endpoint Tox21 QSAR is in-domain for the parent (vancomycin is a training compound). Two cautions: (i) the parent's in-set predictions reflect partial memorisation and are reported instead as measured labels + held-out predictions; (ii) the Tox21 panel covers 12 specific mechanisms only — it does not model vancomycin's clinical effects (nephro-/ototoxicity, Red-Man), which are handled mechanistically and by read-across to the parent's in vivo package.
- **ICH M7 statistical model domain.** The Ames QSAR is out-of-domain for these glycopeptides (Tanimoto $\approx$ 0.22); its negative is low-confidence and the expert rulebase is the primary call. Licensed Derek/Sarah Nexus should confirm in the validated workflow.
- **Conformational sampling.** 3D descriptors/overlaps derive from single low-energy conformers of a large flexible macrocycle → corroborating, not decisive; the decisive evidence is the shared skeleton and conserved toxicophore inventory.
- **Scope.** A hazard-identification / structure-based screening and read-across exercise that supports — does not replace — expert toxicological review and any experimental testing a competent authority may require.
- **Impurity control limits.** Ph. Eur. numerical limits for RRT 0.75/0.87 were not provided; the conclusion assumes control within the applicable monograph limits.

9. References and tools

- ICH Q3A(R2) / Q3B(R2) — Impurities in New Drug Substances/Products.
- ICH M7(R2) — Assessment and Control of DNA-Reactive (Mutagenic) Impurities.
- OECD (2007) — Guidance on the Validation of (Q)SAR Models.
- Benigni R., Bossa C. (2008) — Structural alerts of mutagens and carcinogens (Toxtree/VEGA rulebase; Methodology 1).
- Hansen K. et al. (2009) J. Chem. Inf. Model. 49:2077 — Benchmark data set for Ames mutagenicity (Methodology 2 training data).
- Huang R. et al. — Tox21 / MoleculeNet (Wu Z. et al. 2018, Chem. Sci.) — statistical QSAR training data.
- RDKit (2024) — descriptors, PAINS/Brenk/NIH catalogs, ETKDGV3, MMFF94, Open3DAlign, MCS. Inductive Bio logD/pKa QSAR. Ph. Eur. Vancomycin monograph.
- Sponsor source: "Vanco_RRT_075_and_087_Info" (Xellia) — identities, SMILES/InChI, origin, 91-day dog NOAEL 75/100 mg/kg/day.

Computational screening & structure-based read-across — supports, does not replace, expert toxicological review.